

Selective plant foraging and the top-down suppression of native diversity in a restored prairie

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Summary

1. Clarifying what species are being consumed at what times can improve our understanding of how anthropogenic change affects food web dynamics, with implications for community assembly including restoration. This includes human-based changes to plant communities via species introductions, which can interact with consumer feeding preferences to indirectly alter assembly outcomes including reduced restoration success if planted species are preferentially targeted.

2. We used DNA barcoding of plant material in rodent scat, combined with field-based feeding trials, rodent trapping, and rodent exclosures to test for dietary preferences and assembly impacts of native rodents on a restored and regionally rare tallgrass prairie of central North America. We examined whether native rodents preferred non-targeted and mostly non-native oldfield plants that are more locally abundant, thus protecting the rarer native planted species from consumption, or if rodents preferred native plants regardless of abundance.

3. Our results supported the latter outcome. Barcoding revealed that native rodents consumed mostly non-planted oldfield species for 10 months of the year (92% of the diet). The exceptions were August–September, when planted prairie species accounted for 87% of plants consumed and coincided with their peak seed production. Cafeteria trials suggested diet seasonality to be explained by food limitation – native prairie seeds were consumed year-round when made available. Non-targeted oldfield species thus appeared to be of sufficient quality to support rodent populations (peaking at 23 individuals ha⁻¹ in summer), but without rescuing rarer prairie species from targeted granivory and resulting in reduced prairie diversity outside of experimental exclosures.

4. *Synthesis and applications.* Our work illustrates how anthropogenic-based changes to producer communities may affect feeding pathways in grassland food webs, potentially facilitating establishment by non-target species via indirect consumer effects that can be difficult to detect. These findings suggest that rodents may reduce the restoration success of tallgrass prairie in our region, with some planted species likely to suffer future recruitment difficulties due to granivory. Managers may need to consider multiple trophic levels with restoration, not just the commonly planted producer communities but also the consumers and predators associated with them.

Key-words: diet seasonality, DNA barcoding, granivory, grassland food webs, native rodents, *Peromyscus maniculatus*, plant invasion, *Sorghastrum nutans*, tallgrass prairie, *Zapus hudsonius*

Introduction

A fundamental challenge in quantifying food web dynamics in natural and restored systems is determining consumption patterns of primary producers by consumers, especially what combinations of species are being eaten

and at what times (Pimm, Lawton & Cohen 1991; Martinez 1993; Brown & Gillooly 2003; Kartzinel *et al.* 2015; Tilman & Borer 2015). This question may be especially difficult in anthropogenically transformed terrestrial systems that are targets for restoration, where changes can occur simultaneously to the composition and relative abundance of planted producer communities via plant invasion, and to size and diversity of consumer communities by factors including trophic collapse (e.g. predator

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loss). These systemic transformations of producers and consumers can also interact, as when invasive plants subsidize consumer populations, or when elevated levels of herbivory associated with predator loss transform the composition and diversity of producer communities (Holmgren 2002; Howe, Brown & Zorn-Arnold 2002; Germain *et al.* 2013; Drystek & MacDougall 2014; Kalisz, Spigler & Horvitz 2014). The net effects may alter the assembly outcomes of restored plant communities, in ways that can be difficult to predict or detect.

In grasslands globally, trophic dynamics have been affected by the combined effects of plant invasion, habitat loss, and alterations to the size, diversity, and composition of consumer communities (Parker, Burkepile & Hay 2006; MacDougall *et al.* 2013; Borer *et al.* 2014, 2017). On one hand, these changes underlie the need for grassland restoration in many regions; on the other they create substantial challenges for defining and achieving restoration targets. Of particular note has been the occurrence of so-called 'rodent-plagues' on plant diversity, where small mammal granivory and herbivory drives lost diversity and function including in restored sites (Edwards & Crawley 1999; Howe & Brown 1999; Holmgren 2002; Olofsson *et al.* 2004; Bakker *et al.* 2006; MacDougall & Wilson 2007; MacDougall, Duwyn & Jones 2010; Germain *et al.* 2013; Drystek & MacDougall 2014). These events can unfold in a range of ways depending on the palatability preferences of consumers and whether predation is frequency-dependent, but have been difficult to test due to uncertainties regarding consumer diets. An increase in the frequency of unpalatable invasive plants, for example, can intensify consumer pressure on more palatable native species (Maron & Vilà 2001), while increases in palatable invasive plants can escalate consumption of rarer native flora because consumers are 'subsidized' and become more abundant (Polis, Anderson & Holt 1997; Orrock, Witter & Reichman 2008). Alternatively, high abundance of palatable invasive flora may reduce predation of rarer native flora based simply on frequency (Callaway 2007), while preferential feeding of consumers on palatable invasive plants could facilitate biotic resistance (Parker, Burkepile & Hay 2006). In total, there is considerable support for each of these possibilities although evidence often derives from 'closed' experimental systems using feeding trials or exclosures (Howe *et al.* 2006; Parker, Burkepile & Hay 2006; MacDougall & Wilson 2007). What can be lacking is the testing of dietary preferences in open restored systems (e.g. where feeding preferences are not at least partially influenced by exclosures – Gurevitch *et al.* 1992), how these preferences change seasonally, and the connection between variation in seasonal diet and impacts on restoration success. Seasonality is particularly important given the wide phenological variability in both consumer and producer activity in grasslands (e.g. annual population fluxes by rodents, seed production primarily in late summer or fall – Rabinowitz & Rapp 1980; Boonstra, Krebs & Stenseth 1998; Brady & Slade 2004), with the

possibility of different interaction strengths at different times of the year (e.g., MacDougall, Duwyn & Jones 2010).

Here, we examine these issues by quantifying seasonal variation in diet of a free-ranging native rodent community in an 18 ha restored tallgrass prairie of central North America. The site was cultivated for decades before a species-rich mixture of tallgrass species was planted in 2010 (Germain *et al.* 2013; Drystek & MacDougall 2014). The area surrounding the restored field is dominated by mostly non-native 'oldfield' plants ubiquitous to central North America (Cramer & Hobbs 2007). Some of these non-natives are considered aggressive invaders of North American grasslands (e.g., *Bromus inermis*), as are two non-planted native oldfield species (*Solidago canadensis*, *Conyza canadensis*) that readily invade both farmed and restored areas of the region. Southwestern Ontario once supported ~1000–2000 km² of tallgrass prairie and oak savanna as recently as the early 1800s (Riley 2013). The region is now dominated by settlement, oldfields, and agriculture, with <1% of prairie remaining and most of its species rare and at-risk; restoration of species-rich prairie has become one of the primary conservation tools although success can vary (e.g., Howe 1994). Rodents are known to significantly affect prairie diversity and function in this system (Holmgren 2002), but impacts on large-scale restorations are unknown especially whether rodents may facilitate success by consuming invaders or hamper it by targeting planted native species.

Using molecular techniques combined with field experiments including cafeteria and captive feeding trials we test for seasonal rodent dietary preferences, specifically examining whether rodents favour planted native prairie species, non-targeted exotic and native oldfield species, or both, and how these preferences affect restoration success. We examined whether the rodent diet may be associated more with seasonal variation in plant phenology (e.g. peak feeding during late summer seed production, as is the typical timing of seed set for native prairie plants of this system – Rabinowitz & Rapp 1980), with changes in rodent densities (e.g., high densities in mid-summer typical for our region – Falls, Falls & Fryxell 2007), or some interaction of the two. We then explore the implications for restoring native plant diversity, in terms of impacts on non-targeted flora (i.e. the potential for rodent-based biotic resistance or acceptance) and for the planted native prairie plants that are currently regionally rare (e.g. the potential for increased rarity via rodent consumption, or potential protection via frequency-based rarity advantages).

Materials and methods

STUDY AREA

Our work occurred at the RARE Charitable Scientific Research reserve in the community of Blair in southwestern Ontario,

Canada (43°38.4'N, –80°37.2'W). We focused on two groups of plants – planted native prairie species now rare and considered at-risk regionally, and non-planted oldfield species that are widely distributed regionally including the two native species considered problematic to farmed and restored areas (e.g. *C. canadensis* has evolved resistance to glyphosate). Seed for the planted species in the 18 ha study area was obtained within the region (the Grand River watershed of southwestern Ontario). Specific details on planting can be found in Germain *et al.* (2013).

ESTIMATING RODENT POPULATION SIZE BY SEASON

Rodent live-trapping occurred from December 2011 to December 2012 covering ~1 ha of the prairie study area. Fifty Sherman Live Traps (5 × 7 × 27 cm) were spaced along five 150 m parallel transects that ran from the surrounding invaded area into the planted prairie – 5 traps were in the unplanted area (each 5 m from the planted prairie boundary) and 45 in the prairie. Trapping numbers did not significantly vary by location of the 50 traps, and we routinely recaptured animals in one or the other of the two habitat types, suggesting they were moving back and forth between the two areas (S. Schneider, unpublished data).

The traps were monitored twice daily for 3 days at the start of each month. The traps were removed after each sampling period to avoid habituation. Small mammals were sexed, weighed, and numeric ear tags were placed on the most commonly trapped species (*Peromyscus maniculatus*, *Zapus hudsonius* – see Table 1) in case of recapture. *Myodes gapperi*, *Sorex cinereus*, *Blarina brevicauda*, and *Mustela erminea* were also trapped; the latter presumably entering the traps to predate on captured rodents (two, two, one, and nine captures over the 12 months, respectively – Table 1). All available rodent scat was collected from the traps each month. Due to the movement of animals between the two habitats, we randomly pooled the scat from all individuals into five samples each month (meaning we could not resolve the diet among rodent species or between invaded vs. prairie locations) – these data thus describe the general monthly diet of the entire rodent community (5 samples × 12 months = 60 total samples). Traps were cleaned with tissue and alcohol after each monthly trapping session, to reduce contamination of scat DNA. The scat from the 60 samples was analyzed for plant DNA (see Supporting Information for details), with each plant identified to at least the family level, and

classified as planted native prairie species or non-targeted species, and by functional group [forb, grass, tree, or bryophyte (moss)]. We occasionally detected *Helianthus* sp. [sunflower (*H. annuus*) or woodland sunflower (*Helianthus divaricatus*)] but removed these from the final data set, since the former was used to bait the live-traps while the latter was used in the cafeteria trials and otherwise did not occur on the site.

CAFETERIA TRIALS, EXCLOSURES, AND FEEDING TRIALS

The DNA data indicated that rodents were primarily consuming planted prairie species in 2 months – August and September; diet in the remaining 10 months consisted of mostly non-targeted old-field species occurring in the prairie and in surrounding areas (see Results below). This suggested several possibilities: (i) the rodents prefer non-targeted plants but these are unavailable in these 2 months, (ii) the rodents target that most abundant species, and that native prairie seed is the most abundant at this time, or (iii) the rodents prefer native seed, and it is only available in August–September. We could eliminate the first possibility because most of the non-targeted species flower and set seed at the same times as the native prairie plants (Rabinowitz & Rapp 1980). However, we could not directly separate the latter two, because we lacked seed from the non-targeted species (i.e. the study commenced after these species set seed in 2011). However, we were able to conduct three field experiments with native seed that, in combination with the DNA data and the monthly abundance data for the rodents, allowed us to describe seasonal variation in preference and removal rates of native seed (cafeteria trials), the relative preference for native seed (granivory) vs. seedlings and adult plant parts (herbivory) (feeding trials), and the impact of rodents on recruitment of planted vs. non-targeted oldfield plants in areas lacking rodents (exclosure experiment).

Cafeteria trials

We used monthly cafeteria trials in open field conditions from December 2011 to 2012, with seed of six native herbaceous species [*Rudbeckia hirta* (forb), *Elymus canadensis* (C₃ grass), *Bromus kalmii* (C₃ grass), *Andropogon gerardii* (C₄ grass), *Lespedeza capitata* (forb), *H. divaricatus* (forb)], and a large seeded native tree

Table 1. Total captures of small mammal by live-trapping from December 2011 until November 2012 at the study area. The majority of captures were deer mice and jumping mice. Females largely outnumbered males except for March. Numbers of captured jumping mice peaked in summer; this species hibernates during winter. Deer mice were captured year-round

Month	<i>Peromyscus maniculatus</i> /ave wt (g)	<i>Zapus hudsonius</i> /ave wt (g)	Female/male	Other captures (#)
December	12/15.8	0/0	11/2	<i>Mustela ermine</i> (2)
January	11/14.2	0/0	9/2	–
February	7/16.4	0/0	4/3	–
March	6/14.8	0/0	0/6	–
April	7/23	0/0	5/2	–
May	8/21	1/16	5/4	–
June	10/16.7	4/19.75	7/7	<i>Myodes gapperi</i> (1)
July	16/17.6	7/18.3	14/9	<i>M. gapperi</i> (1) <i>Blarina brevicauda</i> (1)
August	9/14.3	3/14.3	10/2	<i>M. ermine</i> (1)
September	10/16.4	1/17	7/5	<i>M. ermine</i> (5) <i>B. brevicauda</i> (1)
				<i>Sorex cinereus</i> (1)
October	3/18.3	0/0	2/1	<i>Blarina brevicauda</i> (2) <i>S. cinereus</i> (1)
November	3/18.3	0/0	1/2	<i>M. ermine</i> (1)

Acer saccharum (tree). These plants were selected to include four functional groups (tree, C₃ grass, C₄ grass, forbs), to test for variation in group and species-specific seed preferences including whether these preferences varied by rodent population size (Fig. 1). These trials were separated from the trapping area by 500 m. We placed five seeds of each species in separate Petri dishes within a covered plastic container each month, with access only by a 25 cm tube to prevent birds accessing the seed.

Exclosures

We constructed six 4 m² fine-mesh (1 cm) exclosures, buried 50 cm into the ground, to create rodent-free conditions to test recruitment of native prairie plants with and without the presence of rodents (Figs 2 and S1, Supporting Information). This work was factorially combined with a mowing and seed addition treatment ($\pm$ mowing, $\pm$ cage, $\pm$ seed addition = 48 plots), to test the covarying impacts of plant cover and rodents on background recruitment in the prairie vs. recruitment with added seed. Mowing occurred to ground level in March 2012, with all material raked and carried off-site. Plant cover can be critical for rodent feeding behaviour, with open areas potentially magnifying removal because seeds are easier to detect or suppressing removal because of predator concerns (Germain *et al.* 2013). The control for this experiment was an uncaged, unmowed, and unseeded plot on each site. On 8 April 2012, 20 g of a 14-species prairie grass/forb mixture was added to each plot. The prairie grass mixture contained equal numbers of

each seed and was composed of restoration targets including: *R. hirta*, *A. gerardii*, *E. canadensis*, *Oenothera biennis*, *Liatris spicata*, *Poa palustris*, *Penstemon digitalis*, *Sorghastrum nutans*, *Elymis virginiana*, *Schizachyrium scoparium*, *Aster novae-angliae*, *Desmodium canadense*, *Panicum virgatum*, *Monarda fistulosa* (Table S1). During April and May zero planted restoration targets were found in the DNA results, confirming no influence on the DNA analysis. Recruitment success was surveyed in late August 2012. Exclosures were randomly distributed throughout the prairie at least 100 m from the trap lines to try to reduce diet contamination.

Feeding trials

We used 2-week field-based captive feeding trials starting September 2012, to test the relative preference for seeds, seedlings, and adult plants by individual deer mice including how dietary choices changed as supplies of preferred food became exhausted (Fig. 3). Four deer mice were trapped at the far western end of the 50 ha study site, to avoid capturing individuals from the monthly live-trap surveys, and returned to this area at the end of the experiment. One mouse was added to each of four of the fine-mesh exclosures, which were sealed to prevent escape. The cages were stocked with seed from the same species used in the monthly cafeteria trials, plus seedlings of the same species that were grown in a local glasshouse (unheated and unlit) and planted in each cage. Cages were checked daily to count removed seed, and visual evidence of herbivory to seedlings and adult plants.

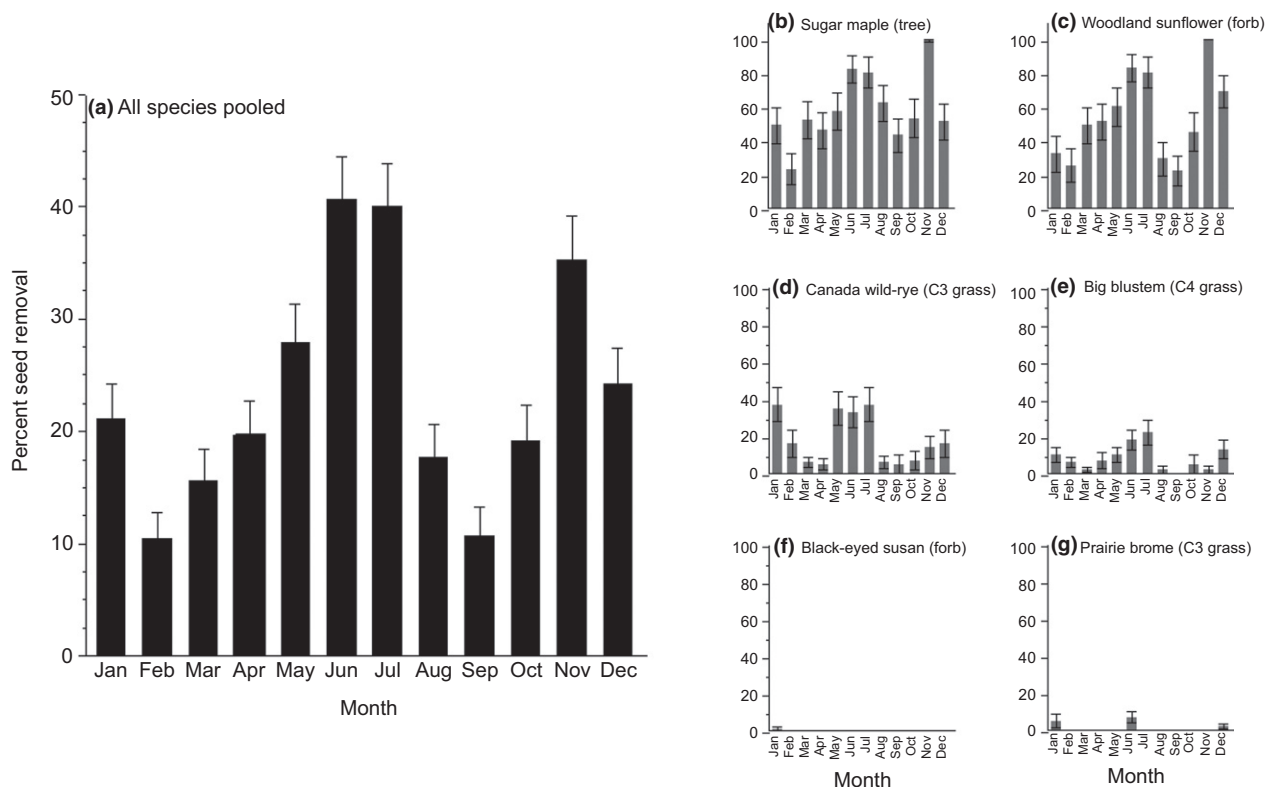


Fig. 1. Percent removal from field seed cafeteria stations by month, showing (a) all species pooled and (b–g) six of the seven native species separately. The grass and forb species are herbaceous species of tallgrass prairie; sugar maple is widespread regionally including to forests adjacent to the study area. The two removal peaks – in June/July and November – corresponded to times of high and low rodent captures respectively (see Fig. 1). Error bars are 1 SE.

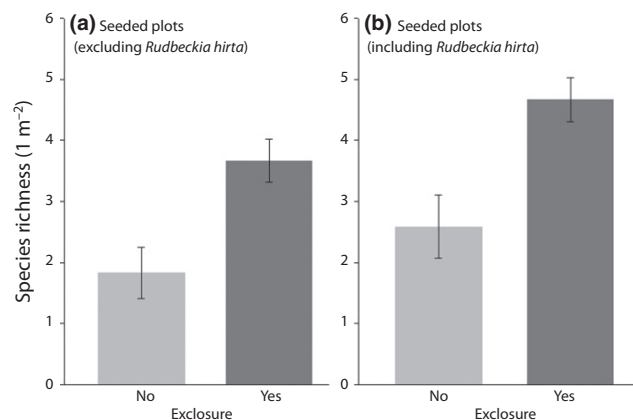


Fig. 2. Increases in species richness per plot following experimental seed additions, with and without the presence of rodent exclosures ($n = 6$ replicates per treatment). Data are shown without *Rudbeckia hirta* (a) and with *R. hirta* (b) seedling recruitment. *Rudbeckia hirta* is a native prairie forb with tiny seeds that does not suffer significant rodent granivory (see Fig. S1) and recruited in large numbers inside and outside of the exclosures. Error bars = 1 SE.

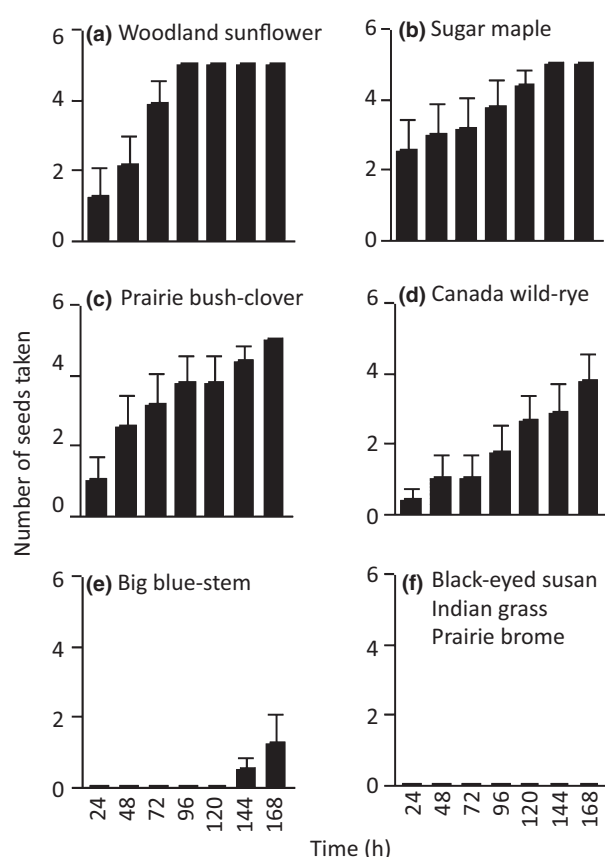


Fig. 3. Seed removal from captive feeding trials using a single deer mouse per cage ($n = 4$ cages). Trials occurred over a 2 week period, although no seed consumption occurred after 168 h. Each plant species (a–f) started with five seeds per cage, with the timing of 100% removal varying by species from the fastest removal (a) *Helianthus divaricatus* (all seeds in all cages gone by 96 h) to the slowest (e) *Andropogon gerardii*. Two species (f) had no seeds removed during the entire trial (*Rudbeckia hirta*, *Bromus kalmii*). Error bars are one SE.

STATISTICAL ANALYSES

Several of our analyses centered on testing month-by-month changes in our measured responses, including temporal variation in rodent density and identity, the identity and total removal rates of seed from the cafeteria trials, how these removals varied

with monthly changes in rodent density, and changes in the proportions of planted prairie vs. non-planted oldfield species in the rodent diet (as determined by DNA analysis). Although we took steps to reduce habituation of rodents to monthly trapping areas and locations of the seed cafeteria trials, we treated these samples as temporally dependent and thus used non-parametric or repeated measures rANOVA (e.g., von Ende 2001). We performed a non-parametric Wilcoxon test to compare monthly variation in rodent density. Using rANOVAs and *post hoc* Tukey HSD tests, we compared the total seed removal and the total number of species consumed from the cafeteria trials by month. Using a linear regression we compared seed removal of *Acer saccharum* to rodent density. Following this we performed a rANOVA and *post hoc* Tukey HSD test to compare the proportion of planted and non-planted species detected in the small mammal diet from the DNA analysis by month. A rANOVA was used to account for scat collection from repeated rodent captures. A linear regression was used to assess the coefficient of determination considering rodent density in comparison to the number of planted and non-planted species detected in the monthly DNA samples. From the captive feeding trials we used rANOVA and *post hoc* Tukey HSD test to compare the number of seeds removed (square root transformed) by species. Also for the captive feeding trial we performed a rANOVA on the categories of seedling damage. No plant damage was detected on adult plants within the cage and no tests performed. Lastly we performed two full factorial ANOVAs using the exclosure experiment comparing the abundance and richness of establishment by planted and unplanted species by treatment. A summary of statistical tests is shown in Tables S2 and S3. Analyses were done using R (v. 3.0.1 – R Foundation for Statistical Computing Vienna, Austria) or JMP (v 10.0.0 – SAS Institute Inc., Cary, NC, USA).

Results

We detected pronounced seasonal diet trends with the monthly DNA sampling, with non-targeted oldfield species constituting most of the diet annually except in late summer when rodents preferentially consumed subsets of native prairie plants presumably as seed ($F_{4,11} = 17.2$, $P < 0.0001$). In total, oldfield plants made up 92% of species in the rodent diet all months except August and September, with planted prairie species accounting for 87% of the rodent diet during that period. There were 38 species detected within the 60 scat samples including

bryophytes (Table S3), with vascular flora representing 26% of all species known to occur in the entire 50 ha study area (i.e. planted and surrounding area – Table S1).

There were 228 rodent captures, with 87 individual animals ear-tagged over the 12 month sampling period. The number of individual captures per ha varied widely by season, as is typical for grassland rodents, with peak densities in mid-summer (July) and prior to the peak in consumption of native plants (August–September) (Fig. 3, Table 1). *Peromyscus maniculatus*, captured every month including during winter, and *Z. hudsonius*, captured from May through September, were the most abundant species. Recaptured animals averaged 2.91 individuals per month (SE = 0.54), and this did not vary seasonally (Wilcoxon test, $P = 0.443$). Nine *M. erminea* were captured in the latter months of the year (December 2011 and August–November 2012), and were correlated with a sixfold reduction of deer mouse population size during August and September 2012 (Table 1). Six *M. erminea* captures occurred in traps that also contained a mutilated mouse carcass.

There were varying relationships between the density of the rodent community by month and the rodent diet as revealed by the DNA analysis (Fig. 4). Rodent density per month and numbers of oldfield species consumed per month were positively correlated ($R^2 = 0.166$, $P = 0.041$), while no relationship was detectable between rodent density and native plant consumption ($R^2 = 0.0786$, $P = 0.43$). The latter result was explained by native plant consumption at both high and low rodent densities (July and November respectively). No planted native prairie species were observed in the scat samples in most months of the winter and spring, suggesting that seeds released during autumn were completely consumed, harder to detect by rodents at those times, or that the rodents had switched their preferences to non-native species. This result indicates that rodents are not seed caching native prairie plants.

Seed removals from the cafeteria trials occurred in every month, with no seasonal variation in species preference based on order of removal from the feeding stations (i.e. the same subset of species were always consumed first

– Fig. 1). In all months, rodents targeted the same four species: a tree (*A. saccharum*), two large-seeded native forbs (*H. divaricatus* and *L. capitata*) and a native C_3 grass (*E. canadensis*). *Andropogon gerardii*, which was ignored most months, was consumed during June and July when rodent densities were highest, while the remaining two species (*B. kalmii* and *R. hirta*) were almost never taken ($F_{11,156} = 2.039$, $P = 0.0281$ Tukey's HSD). There were variations among months in the amount of seed taken – removals from the cafeteria stations dropped significantly in late summer ($F_{11,156} = 2.013$, $P = 0.0305$ Tukey's HSD), which corresponded with the peak in seed production in the field, when the planted species were most prevalent in DNA scat, but also when rodent captures declined (Fig. 4). There was no correlation between rodent densities by month and the removal of preferred seed from the feeding stations – removal was similarly high in summer during the population peak and when rodent numbers were low in winter ($R^2 = 0.210$, $P = 0.137$).

The captive feeding trials demonstrated the identical order of seed consumption as observed in the field, with preferences for the same four species (Fig. 3, $F_{3,7} = 117$, $P < 0.0001$, Tukey's HSD). Three of the seven species had their seeds entirely consumed after 5 days (Fig. 4). Seedling herbivory was only observed after all preferred seeds were taken, and only for two Asteraceae forb species out of the seven planted species ($F_{3,7} = 16.6$, $P < 0.0001$). We detected no consumption of plant parts of adult native species in our captive feeding trials.

The exclosed plots had twice the abundance of planted native seedlings ($F_{1,5} = 5.92$, $P = 0.0196$) and significantly higher richness of planted species ($F_{1,5} = 13.8$, $P < 0.001$) than areas lacking protection from small mammals (Figs 2 and S1). Most notable was *E. canadensis* (C_3 grass), with 51 individuals observed in the exclosures, but none in open areas. There were no recruitment differences for oldfield plants inside vs. outside the exclosures. The richness of planted prairie species outside the cages was especially low in seeded plots that were experimentally defoliated with mowing (Fig. S1), suggesting that small mammals were more able to detect native seed in these

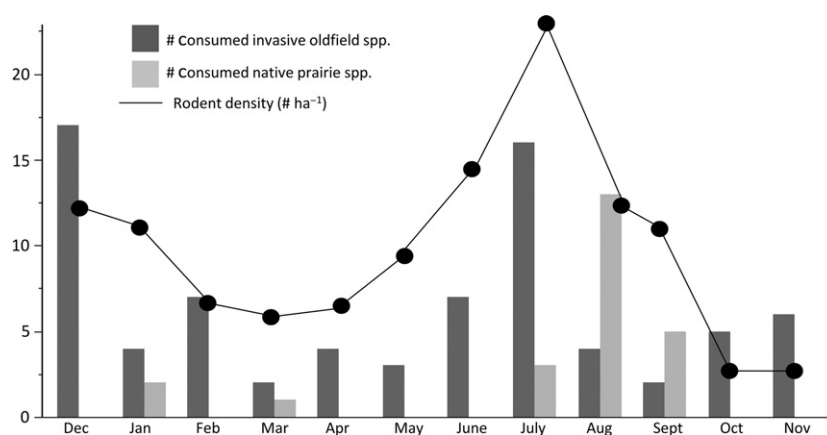


Fig. 4. Variation in the density of live-trapped rodents (# of individuals per hectare per month) and the number of consumed native prairie and invasive plant species per month at the restored prairie 2011–2012. The y-axis represents both rodent density and the number of consumed species.

areas – species richness was reduced on average $4.6 \times$ compared to the same treatment in the cages ($F_{1,47} = 7.23$, $P = 0.0104$ Tukey's HSD).

Discussion

We tested for seasonal dietary preferences of native consumers in a restored and anthropogenically transformed plant community, where formerly widespread native plants are now isolated with a regional matrix of introduced and native oldfield species including some considered invasive to North American native grassland. We specifically examined the relative preferences of native rodent consumers for prairie vs. non-targeted and regionally widespread oldfield plants under open conditions (i.e. free-ranging animals), whether these preferences changed seasonally, the potential impacts on the diversity of one or both plant communities, and how this affected restoration success several years after planting. Past work on anthropogenic-based changes to resource–consumer interactions (i.e. combinations of native and non-native producers and consumers, which is increasingly common globally) has suggested a range of possible outcomes (Parker, Burkepile & Hay 2006), which can be hard to test given the difficulty in tracking consumption patterns of free ranging individuals and whether these patterns change by season (Tilman & Borer 2015). These outcomes can include native consumers preferring native plants despite their increasing rarity, native consumers relying on non-native plants given their prevalence regionally, or no consumer preferences if the resident species happen to be generalist and opportunistic feeders (e.g. Mack & Thompson 1982; Futuyma & Moreno 1988; Thompson 2005; Parker, Burkepile & Hay 2006; Borer *et al.* 2014).

We mostly observed the first outcome, with native consumers showing preferences for subsets of planted native prairie species despite their rarity regionally, and almost exclusively at times of peak seed output. The consumption of seeds of these plants were observed year-round in the monthly cafeteria trials, with same subset of species taken consistently and in the same order, suggesting that consumption levels in this system could be determined more by seed availability than by changes in consumer preference by season (i.e. seed consumption appears to end in late fall because of food limitation, rather than switching to different species). At other times of the year, the rodent community diet consisted of a wide array of oldfield plants, some that are invasive to restored prairie and occur widely within the region, with no indication, based on the DNA evidence, of herbivory on adult native prairie plants or the caching of their seed.

We cannot comment on the diet of the rodent community prior to the arrival of exotic plants in the 19th century, or how the present-day diet would change if native plant populations were more widespread. However, with our observed deer mouse population densities at normal to high levels for southern Canada (Davidson & Morris

2001), the implication is that these introduced species provide sufficient resources to maintain rodent populations in this system. If true, the net effect is likely twofold: normal to high population levels of rodents being maintained, which in turn exerts intense granivore pressure thereby inhibiting recruitment by some native prairie plants. While the latter observation has been observed previously (e.g. Howe, Brown & Zorn-Arnold 2002; MacDougall & Wilson 2007), its causal connection with rodent feeding preferences in systems with widespread introduced plant populations has not been directly demonstrated because of the difficulty in assessing diet without DNA methods. This targeted granivory may present a substantial challenge for the restoration success of planted high-diversity prairie, if subsets of species are preferentially consumed by native rodents once the prairie reaches canopy closure and rodent populations become established. This creates a subtle but difficult restoration challenge – prairie plantings on unvegetated fields may initially succeed because rodents are absent (see Germain *et al.* 2013), but future recruitment may be significantly constrained for some species.

Co-evolved linkages between consumers and producers have been hypothesized to be important for determining changes in plant community composition with human disturbance, potentially driving invasion meltdowns or, conversely, biotic resistance (Parker, Burkepile & Hay 2006; Pearse & Altermatt 2013). We observed high consumption levels of native seed by native consumers, suggesting the possible facilitation of introduced species initially – if natives plants were preferred, all things being equal, this would have provided fitness advantages for establishing invasive populations of incoming oldfield species introduced accidentally or intentionally by humans (MacDougall, Gilbert & Levine 2009; Kalisz, Spigler & Horvitz 2014). However, we also see that these regionally prevalent oldfield species now constitute the bulk of the rodent diet in most months. The connections between consumer diet and the provenance of producer species, therefore, were relatively weak compared to some previous predictions that coevolved specializations might be strongly conserved (e.g. the sole dependence of native rodents on native plants) in food webs (Pearse & Altermatt 2013). This result may derive partly by deer mice – the dominant rodent species in our system – being a generalist and sometimes opportunistic feeder (Davidson & Morris 2001). It may also relate to oldfield plants, introduced but also including native species such as Canada goldenrod, now serving as the main standing carbohydrate reserve on the landscape, given the dominance of these plants across the region for at least the last century – it seems likely that there would have been strong pressure on rodents to utilize these flora. This reliance on mostly non-native oldfield plants plays an unclear role in the current regulation of these plant populations, in our restored prairie and more generally within our study region. One implication is the expansion rate of these plant species, including

some aggressive invasives, could even be curtailed by rodent consumers, compared to what it would be in their absence (e.g. Levine, Adler & Yelenik 2004).

The DNA data provided insight into consumer–producer interactions in restored grassland that would be otherwise difficult to detect, including seasonal variation in rodent diet and the potential for indirect consumer effects on plant interactions (Holmgren 2002). For example, at peak rodent population density in July we observed the widest range of plant species in the rodent diet, including non-native and potentially invasive trees, grasses, and forbs, as well as the most rapid depletion of seed from the widest range of plants in the cafeteria trials – both suggest the possibility of density-related responses to food limitation, which can be fierce in deer mouse populations (Davidson & Morris 2001). Rodent densities subsequently dropped by 54% in the next month, despite peak seed production by preferred native species, and remained in decline for the rest of the year. There could also be a predatory influence of weasels, which were trapped in highest numbers in late summer. Weasels can have rapid and sizeable impacts on grassland rodent populations (Hanski *et al.* 2001), although the reduced rodent population size was not associated with changes in the magnitude of seed removal from the cafeteria trials. This accidental observation of weasel predation does raise an important consideration for grassland restoration – the emphasis tends to be on plants species with an assumed ‘build it and they will come’ model for associated insects, herbivores and predators. As our work suggests, not all of these groups may re-appear in similar abundance or at similar times, with the potential for strong and unanticipated effects such as selective rodent granivory.

Our most striking finding, deriving largely from the DNA data, was the strong seasonal switching of plant consumption between planted vs. oldfield plants, and its implications for how consumer–producer interactions shape restoration outcomes in our system. The consumption of non-targeted oldfield species appears sufficient to maintain rodent populations at high levels (Davidson & Morris 2001). However, this seems to be insufficient to shelter rarer prairie species from targeted granivory, which in turn may hamper their persistence. This observation is consistent with the demonstration of indirect consumer effects influencing shifts in invasive-native plant abundances in grasslands generally (Parker, Burkepile & Hay 2006; Pearson & Callaway 2008). It also suggests that some prairie plants, already regionally rare, may be locked in an extinction debt that may be difficult to solve with seed additions alone, with little consumption detected on adults or seedlings (Germain *et al.* 2013) but intense granivory that suggest future recruitment difficulties (Bricker, Pearson & Maron 2010). The occurrence of such so-called ‘rodent plagues’ (Howe *et al.* 2002) thus presents a significant restoration challenge that may be

difficult to manage, especially if rodents are native, regionally widespread, and have impacts difficult to observe (i.e. ‘cryptic consumers’ – Keesing 2000). Some work suggests that maximizing the size of restored prairie areas may increase the odds of plants escaping detection by rodents during recruitment (Germain *et al.* 2013). It may also be important to consider restoring habitat features that favor rodent predators such as raptors and small mammalian carnivores such as badgers (*Taxidea taxus*), although these species tend to be even rarer and more at-risk in grasslands than the plants they may indirectly protect (e.g. Herriot 2009; Ethier *et al.* 2012).

Authors' contributions

S.S. and A.S.M. designed the research, all field work was conducted by S.S., R.S. and S.N. did the DNA analytical work, and each author helped write and edit the paper.

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Data accessibility

All novel sequences generated in this study have been deposited in the Barcode of Life Database, as well as GenBank (see Tables S4–S6). All data have been archived in Dryad Digital Repository <https://doi.org/10.5061/dryad.d3m23> (Schneider *et al.* 2017).

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Supporting Information

Details of electronic Supporting Information are provided below.

Fig. S1. Contrast in native prairie plant recruitment, between seeded and uncaged plots with no vegetation cover, vs. seeded and caged areas with no vegetation cover.

Table S1. Total list of species recorded from the study area.

Table S2. Summary of statistical tests.

Table S3. Detailed outcome of the full-factorial ANOVA of the enclosure experiment.

Table S4. Monthly occurrences of plants detected by the DNA analyses.

Table S5. Taxonomic determinations, Barcode of Life Database Sample ID #'s, Barcode of Life Database Process ID numbers and GenBank accession numbers.

Table S6. Supplemental methods – DNA barcoding analysis.